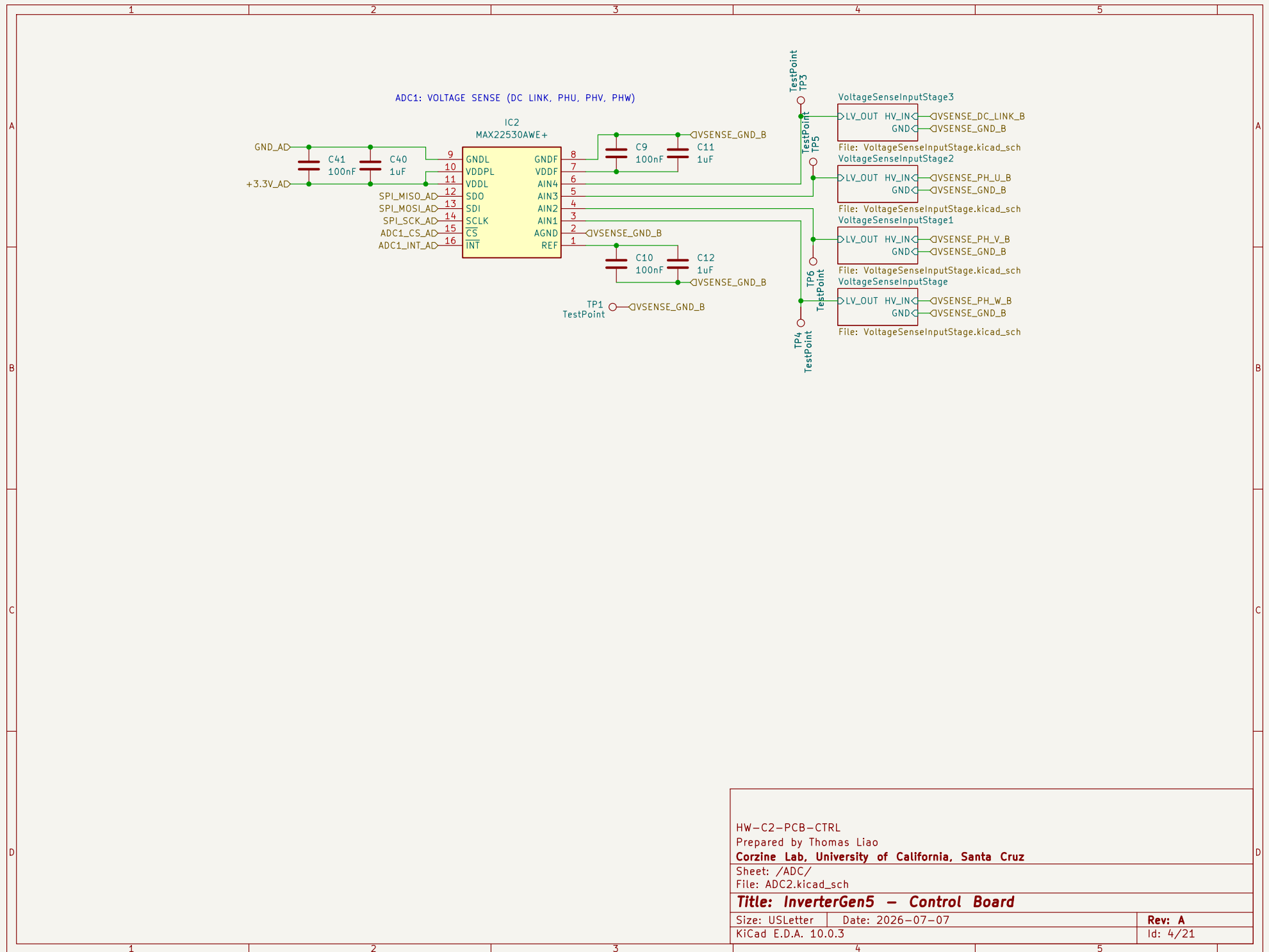
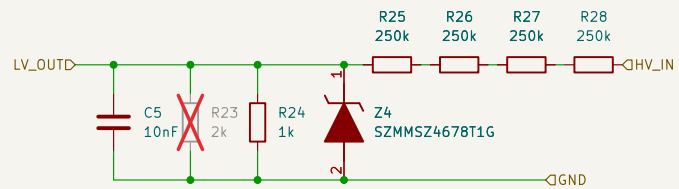


NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,  
DISABLE TIM1 SYNC PIN (TIM1 ETR)



Rev: A  
Id: 1/21





ZENER CLAMPS ADC INPUT AT 1.8V  
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad\_sch

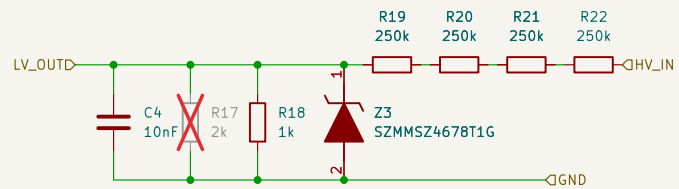
**Title: InverterGen5 - Control Board**

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V  
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad\_sch

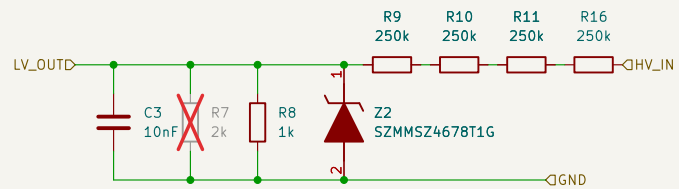
**Title: InverterGen5 - Control Board**

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V  
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad\_sch

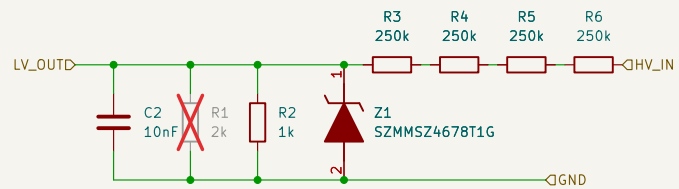
**Title: InverterGen5 - Control Board**

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V  
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad\_sch

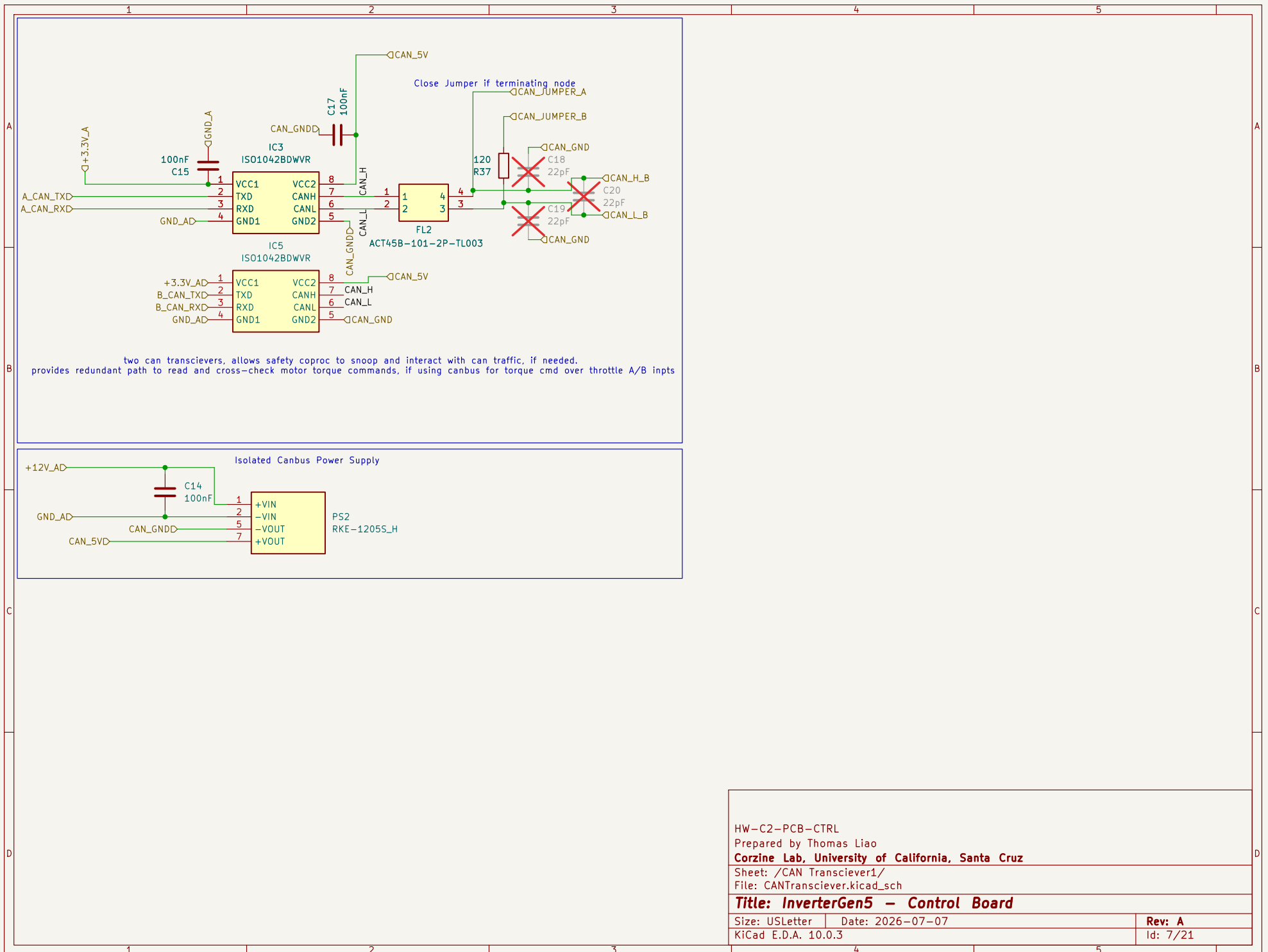
**Title: InverterGen5 - Control Board**

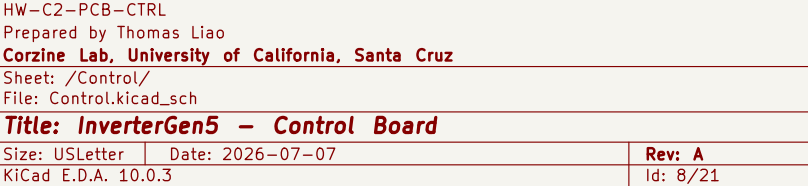
Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

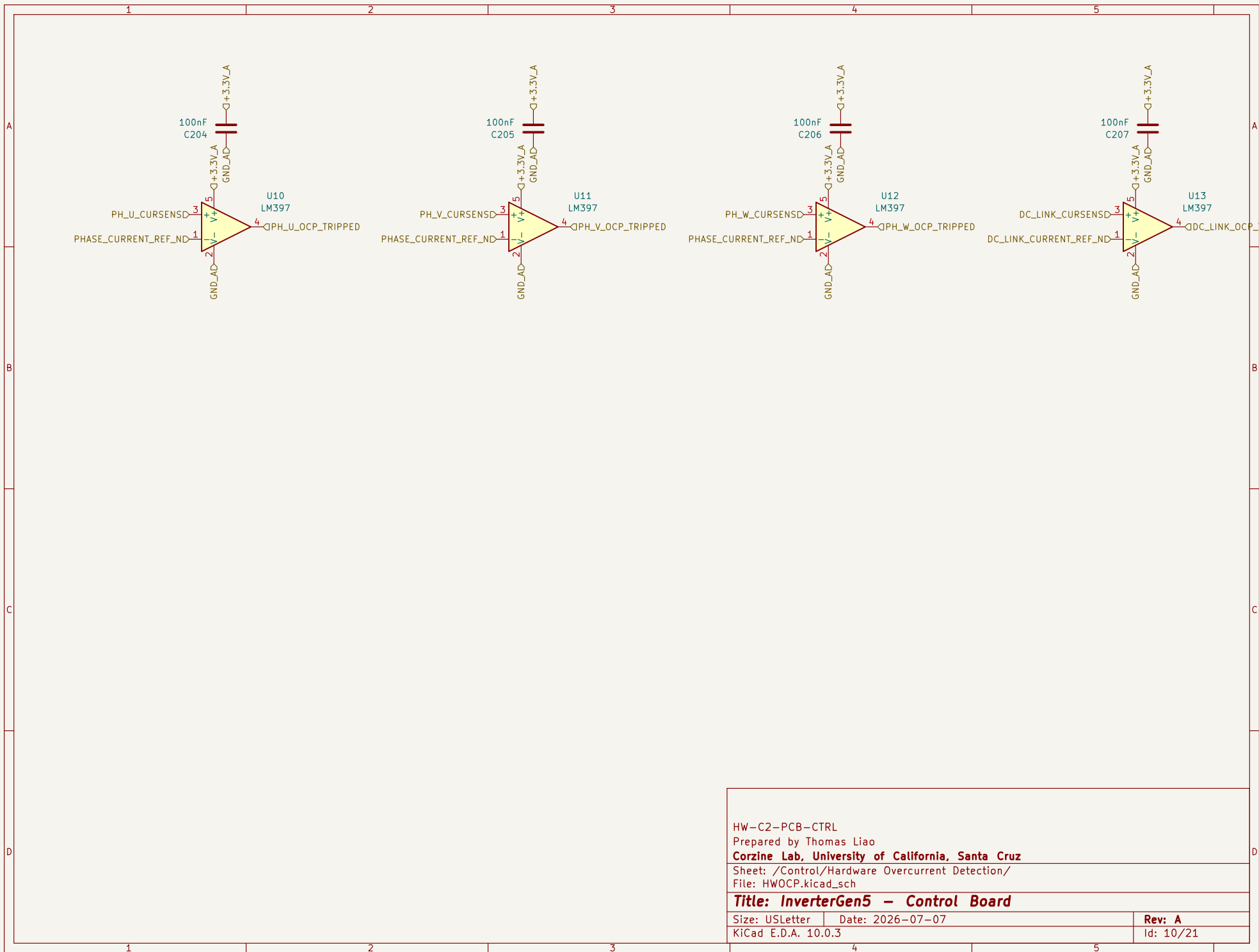
Rev: A

Id: 6/21









HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad\_sch

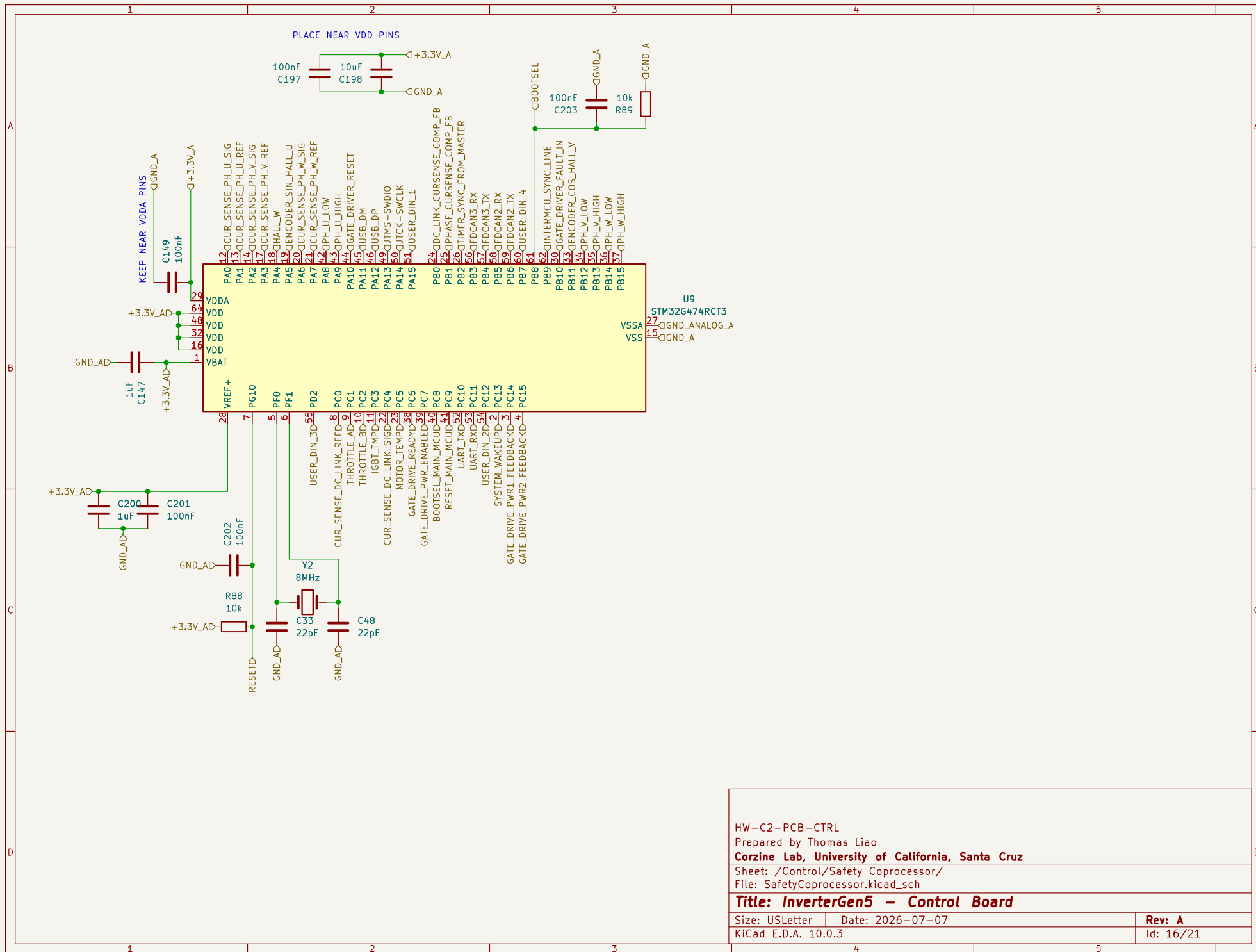
**Title: InverterGen5 - Control Board**

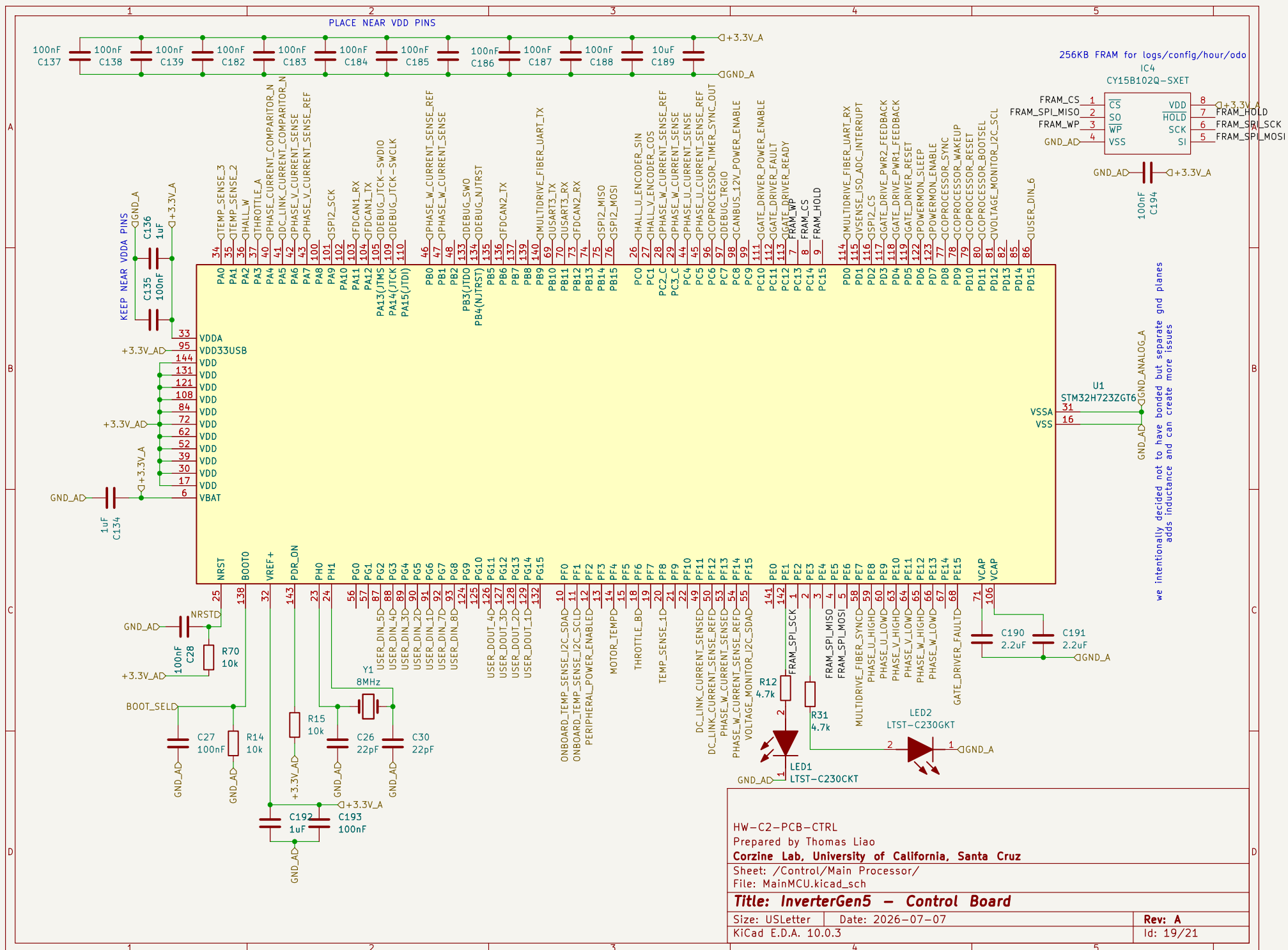
Size: USLetter Date: 2026-07-07

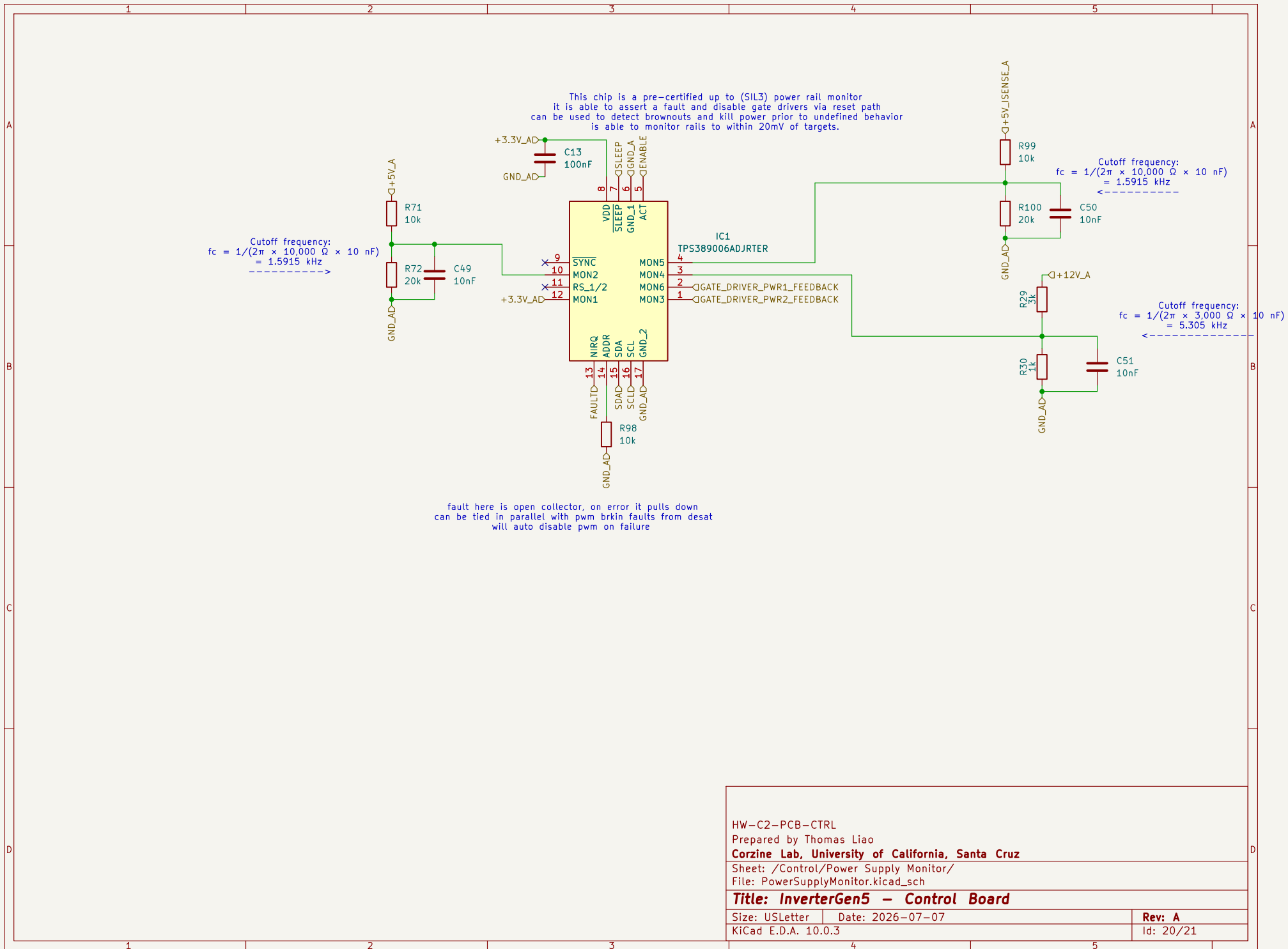
KiCad E.D.A. 10.0.3

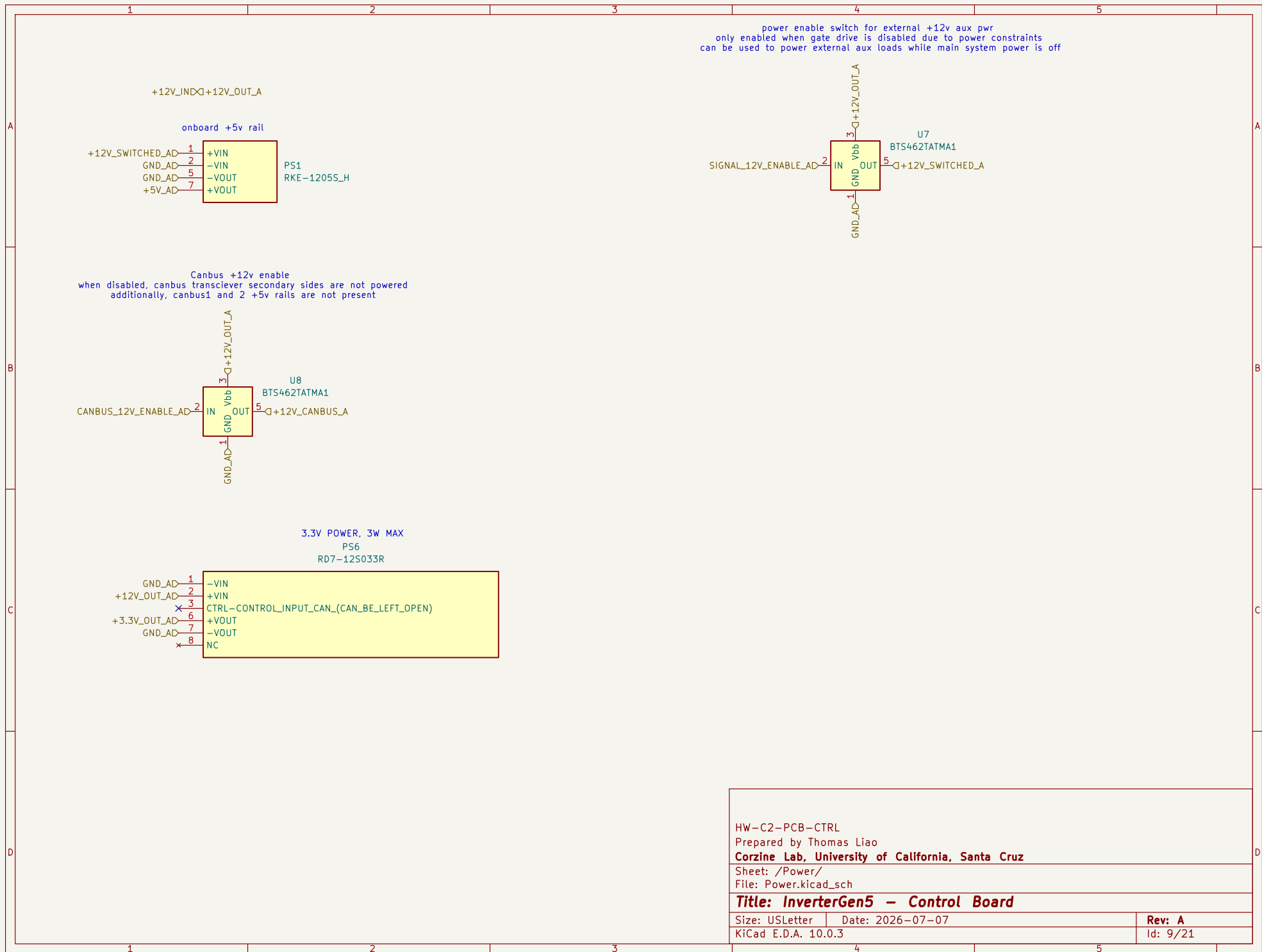
Rev: A

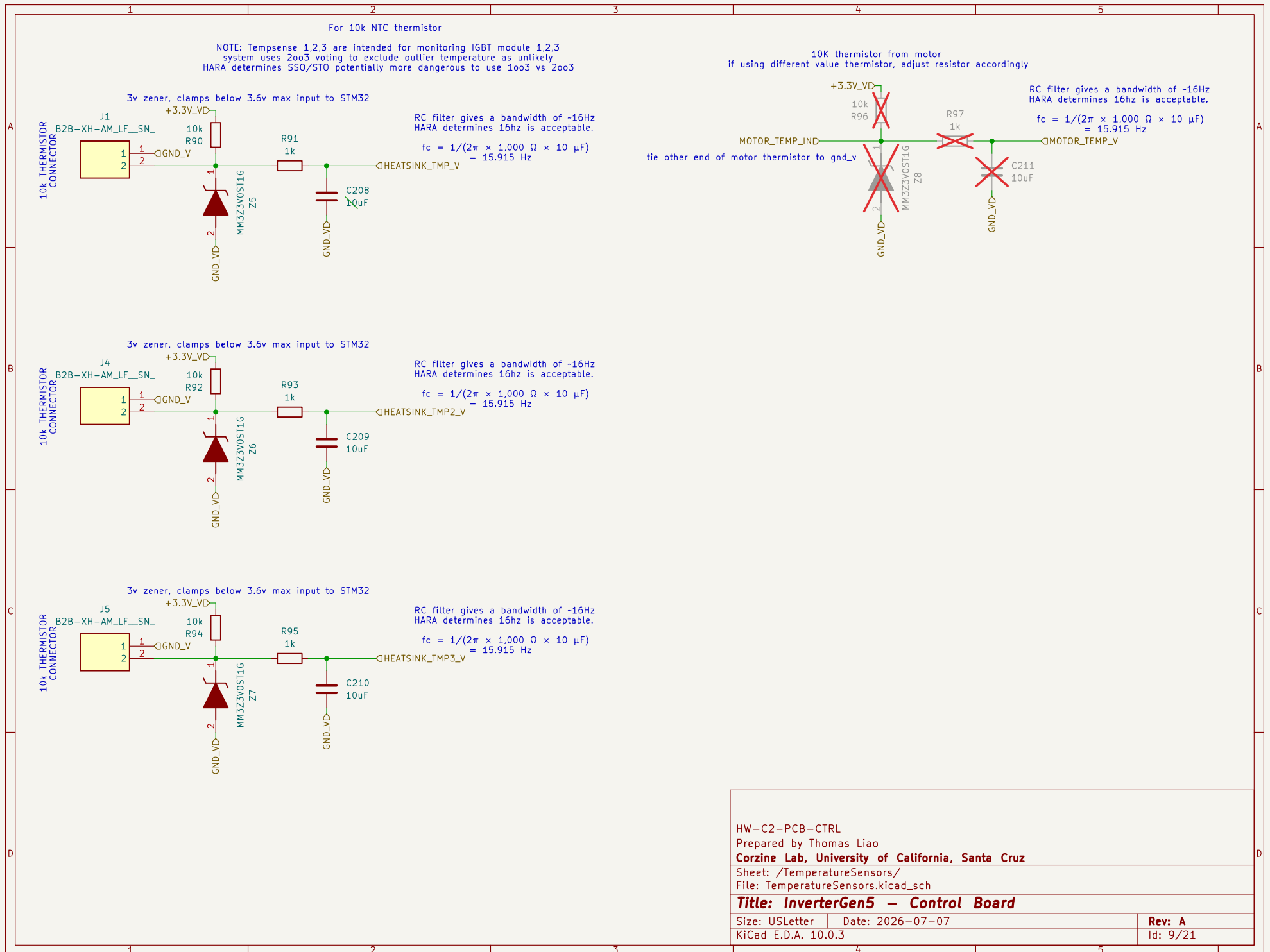
Id: 10/21



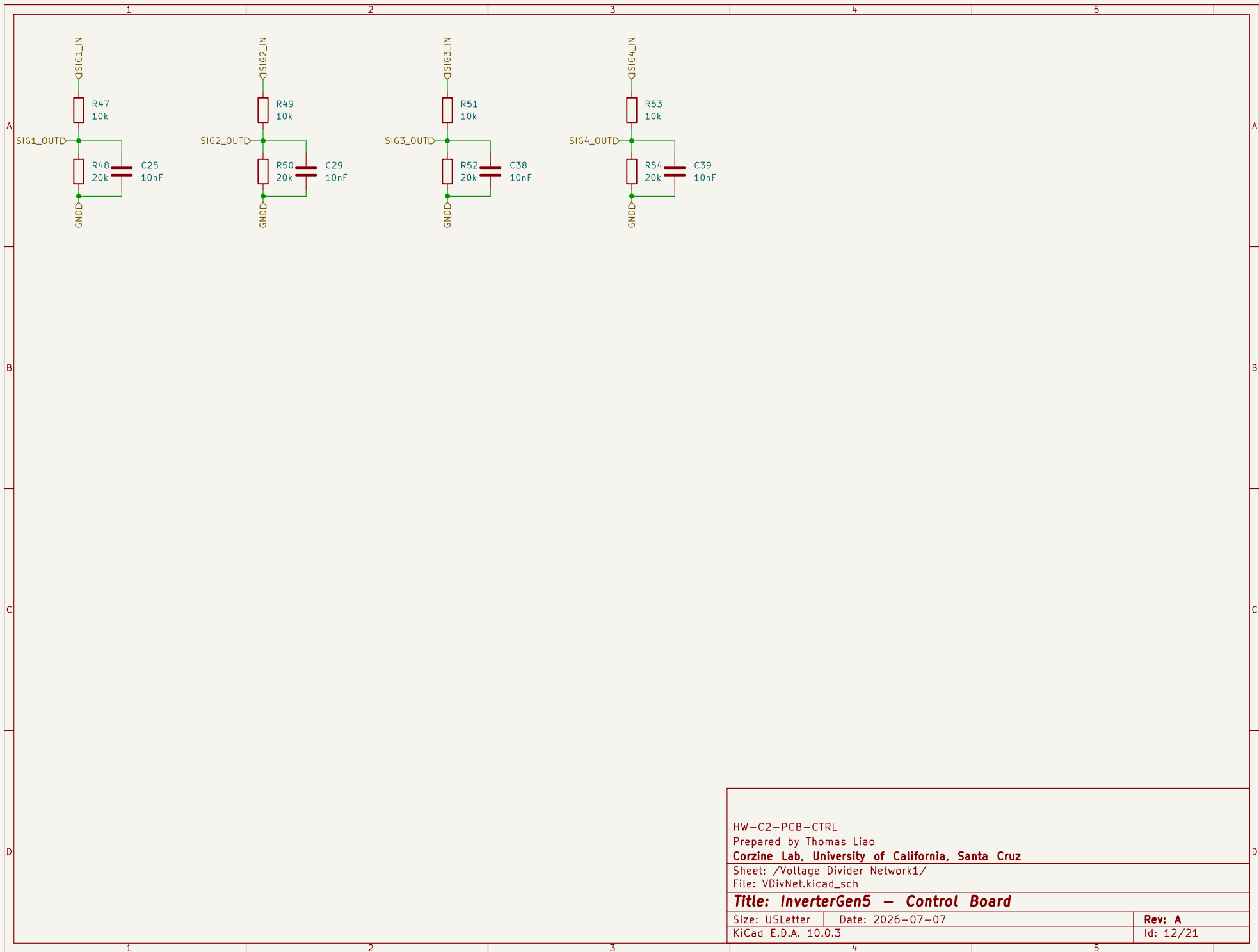












HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad\_sch

**Title: InverterGen5 - Control Board**

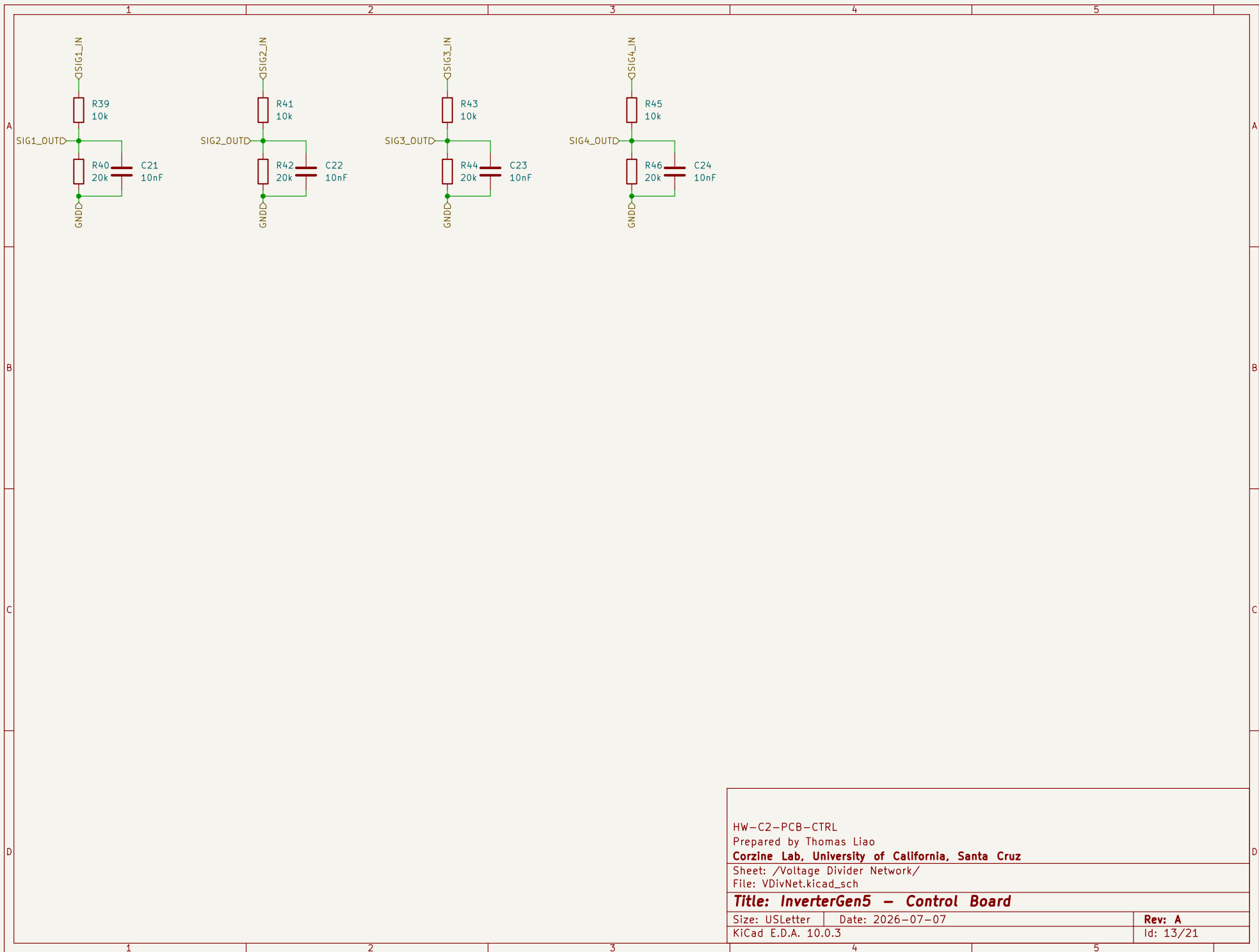
Size: USLetter Date: 2026-07-07

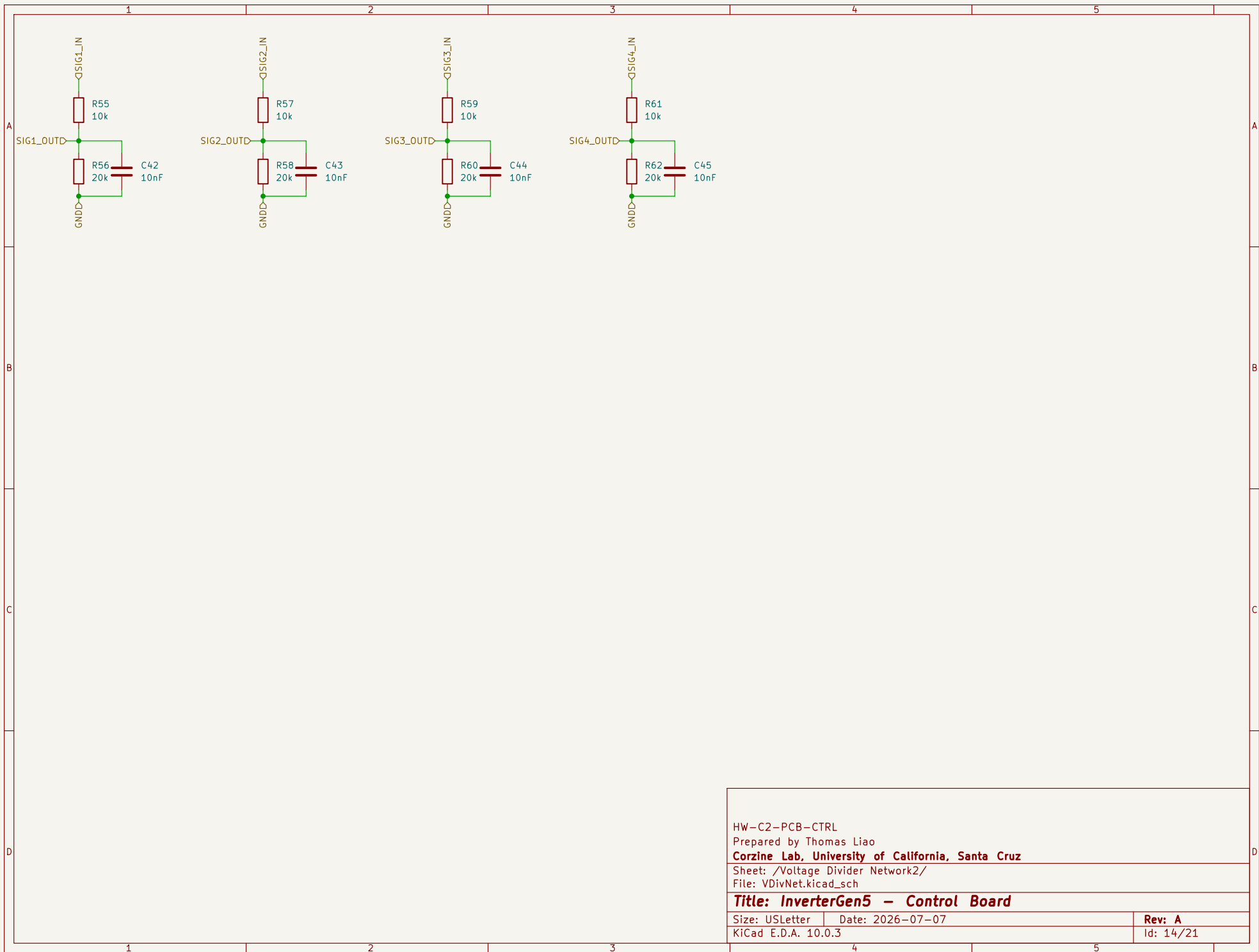
KiCad E.D.A. 10.0.3

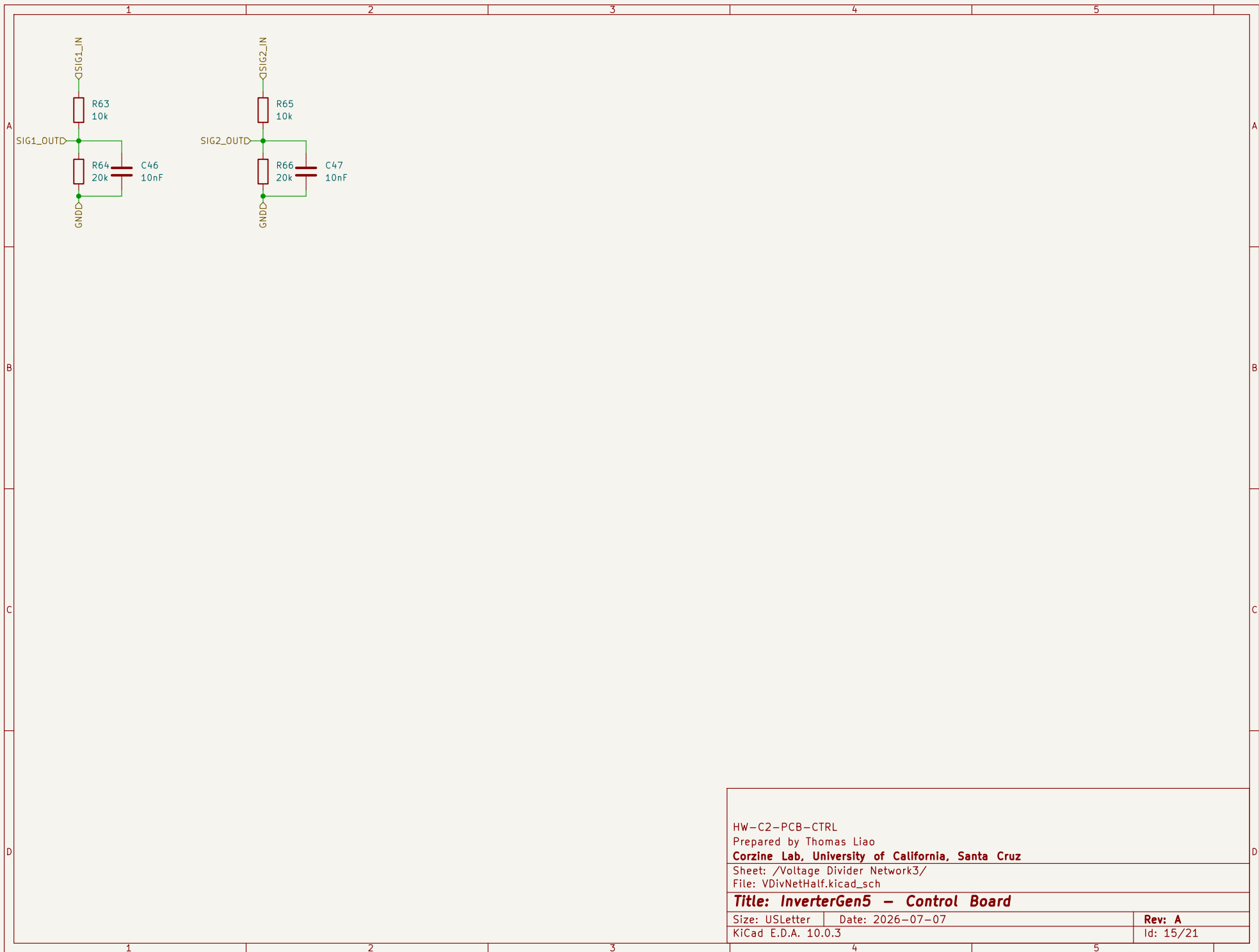
Rev: A

Id: 12/21









|                                                   |                  |           |
|---------------------------------------------------|------------------|-----------|
| HW-C2-PCB-CTRL                                    |                  |           |
| Prepared by Thomas Liao                           |                  |           |
| Corzine Lab, University of California, Santa Cruz |                  |           |
| Sheet: /Voltage Divider Network3/                 |                  |           |
| File: VDivNetHalf.kicad_sch                       |                  |           |
| Title: InverterGen5 - Control Board               |                  |           |
| Size: USLetter                                    | Date: 2026-07-07 | Rev: A    |
| KiCad E.D.A. 10.0.3                               |                  | Id: 15/21 |

